QUALITATIVE ANALYSIS FOR THE SCENARIO OF *SLEEC@run.time*

To validate the SLEEC rules we elicited for the assistive robot scenario, we conducted semi-structured interviews with domain experts. Specifically, the validation process focused on the following validation question (VQ):

> **VQ.** Do the elicited SLEEC ruleset, their connected principles, and associated SLEEC labels provide a *clear*, *complete*, *correct*, and *value-aligned* representation of ethical considerations with respect to the reference scenario?

Methodology and domain experts' selection

We followed the *qualitative survey empirical standard*[1] and its essential attributes to perform semi-structured interviews. Participants were selected through *convenience sampling*[2], commonly used in qualitative research, to gather expert feedback.
Convenience sampling is a non-probability sampling method where individuals are selected based on their accessibility and availability to the researcher. This approach focuses on participants who are easily reachable and willing to take part in the validation, including colleagues, nearby professionals, and people already in contact with our research team.
Accordingly, we contacted several experts.
The criteria used for recruiting them were based on their knowledge and expertise with SLEEC rules. Specifically, each has contributed research publications in the fields of software engineering, ethics-based systems, digital ethics, AI, and focusing on SLEEC-based systems.
We eventually interviewed four domain experts.

Two of the authors conducted synchronous, one-on-one conversations with individual experts, where each conversation was driven by an interview protocol.
They introduced through a brief presentation outlining the study's objectives, the reference scenario, and background concepts, followed by a discussion structured around a set of open-ended questions. Each interview lasted between 1 and 1.5 hours.
Participants' answers were audio-recorded and used to apply qualitative data analysis, focusing on the relevance, expressiveness, and coherence of answers in relation to the study's objectives.

Experts' answers were used to iteratively adjust the ruleset. Suggested changes were incorporated once all authors agreed and any issues of clarity or practical applicability were addressed. The final SLEEC ruleset was used in the implementation and is available in the *Final SLEEC ruleset.pdf* document.

[1] https://github.com/acmsigsoft/EmpiricalStandards/blob/master/docs/standards/QualitativeSurveys.md
[2] Samuel J Stratton. 2021. Population research: convenience sampling strategies. Prehospital and disaster Medicine 36, 4 (2021), 373–374.

We following present the detailed profiles of the experts, while the transcripts of all four interviews are available in the *IDx-Interview Transcription.pdf* documents.

Profiles of the experts

[id:1] The expert holds a PhD in Artificial Intelligence and is currently an assistant professor in computer science. Their research focuses on the application of logic-based approaches within Artificial Intelligence, with interests spanning multi-agent systems, ontologies, games, social choice, and formal philosophy. They have extensive academic experience across several European institutions and have contributed to both teaching and research in computer science and knowledge representation.

[id:2] The expert is a PhD candidate in Computer Science. They hold a master's degree in Data Science and a bachelor's degree in Computer Engineering with a focus on Software Engineering. Their current research interests lie in the areas of software testing and verification of complex systems of ethical-aware autonomous systems.

[id:3] The expert holds a PhD in Computer Science and currently serves as a faculty member in the field. Their research focuses on applying logic-based approaches to the formal verification of complex and safety-critical systems, with a particular emphasis on software verification. They have a strong background in formal methods, including extensions of temporal logic and reasoning for multi-agent systems. Their work often involves the development of practical tools that support theoretical research and are made publicly available. Their broader research interests include Artificial Intelligence, Software Engineering, Logic, and Formal Verification. Before taking on their current role, the expert spent over two decades in academic and industry-related research positions.

[id:4] The expert holds a PhD in Information and Communication Technology, currently serving as an assistant professor, with a background in privacy-enhancing technologies, mobile applications, and empirical software engineering. The research they are conducting focuses on designing and evaluating systems that prioritize user privacy, particularly in mobile contexts, through the development of innovative technical solutions. They approach the topic of ethical-aware systems from a human-computer interaction perspective, emphasizing the role of user experience and behavior in their design.